

Student Name: Sivasuthan S.

Student No: 258826B

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Module Name: Semantic Web and Ontological Engineering

Ontology-Based Smart Housing & Rental Finder for Sri Lanka

This project aims to develop a smart housing and rental finder designed specifically for Sri Lanka. It uses semantic web technologies to organize important property information such as location, rental price, available facilities, and user preferences. Through the use of ontologies, knowledge graphs, and SPARQL queries, the system can identify connections between different property details and help users find suitable houses or rental properties based on their needs.

1. Problem Statement

Finding a suitable house or rental property in Sri Lanka can be difficult because property information is not always organized in a clear and structured way. Many existing property websites mainly depend on simple keyword searches, which may not always give users the results they are looking for. Users can also face difficulties when trying to apply detailed filters or find properties that closely match their individual preferences.

Current rental platforms suffer from:

- Limited understanding of relationships between tenants, properties, locations, and rental agreements.
- Limited intelligent filtering, making it difficult to find properties based on connected concepts and specific requirements.
- No clear visual view of relationships between different housing-related entities.
- Less relevant search results, as properties may not be ranked according to the user's actual preferences.
- Limited support for future intelligent features, making it difficult to expand the system for advanced recommendations and decision-making.

2. Proposed Solution

This project aims to develop a web-based housing and rental finder for Sri Lanka using an ontology-based approach. The system organizes housing-related knowledge using RDF and OWL, allowing different property details to be linked and represented in a structured way. This makes it easier for users to search and find relevant housing information based on their requirements.

The screenshot shows a web browser window with the title 'Sri Lankan Housing Finder'. The address bar shows '127.0.0.1:5000'. The page has a header 'HOUSING & RENTAL MATCHING' and a main heading 'Find your next home in Sri Lanka'. Below the heading, there is a search form with two input fields: 'TENANT TYPE' (set to 'Any') and 'CITY' (set to 'Colombo'). A 'Search' button is to the right. Below the search form, a 'BEST CHOICE' section highlights 'Room_Colombo_Student_001 — Colombo'. Below this, a 'MATCHES' section displays a table of results.

TENANT	PROPERTY	CITY	SCORE
Student	Room_Colombo_Student_001	Colombo	4
Student	Apartment001	Colombo	2

The system models:

- Property (Apartment, House, Room)
- City
- Tenant
- Lease
- Landlord

Key relationships:

locatedIn (Property → City)

leaseFor (Lease → Property)

ownedBy (Property → Landlord)

The system provides:

- A web interface for searching properties
- Intelligent result ranking
- Highlighting of the best choice
- A dynamically generated knowledge graph
- Ontology upload support

3. Technologies Used

Ontology and Semantic Technologies:

- OWL (Web Ontology Language)
- RDF (Resource Description Framework)
- RDFLib (Python)

Backend Technologies:

- Python
- Flask

Query and Reasoning Technologies:

- SPARQL
- Soft semantic filtering and ranking

Knowledge Graph Visualization:

- PyVis

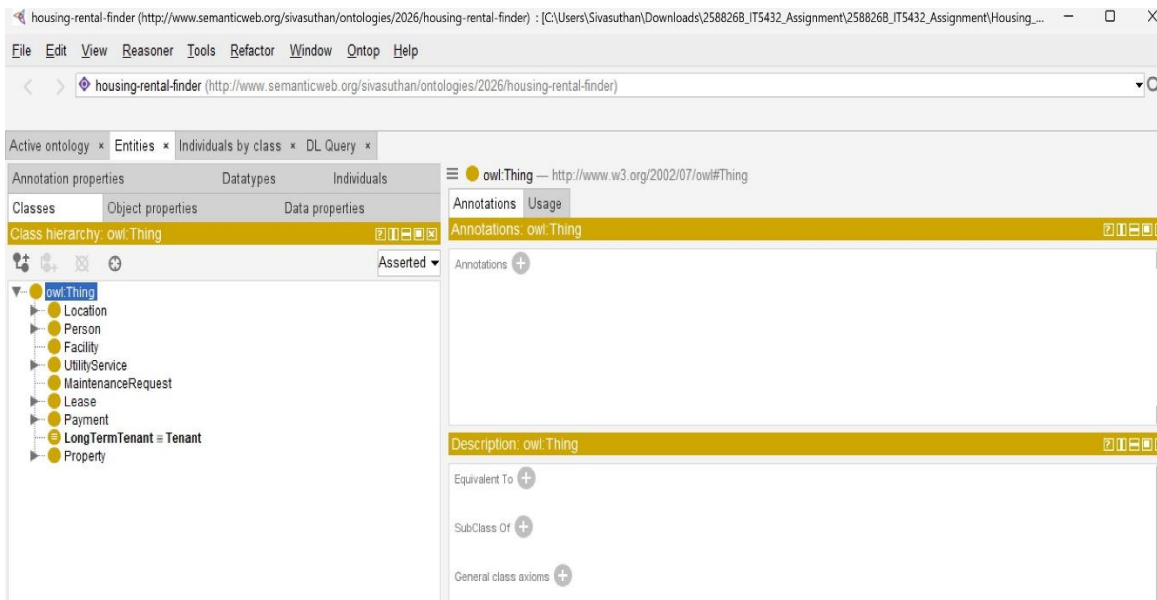
Frontend Technologies:

- HTML5
- Bootstrap 5

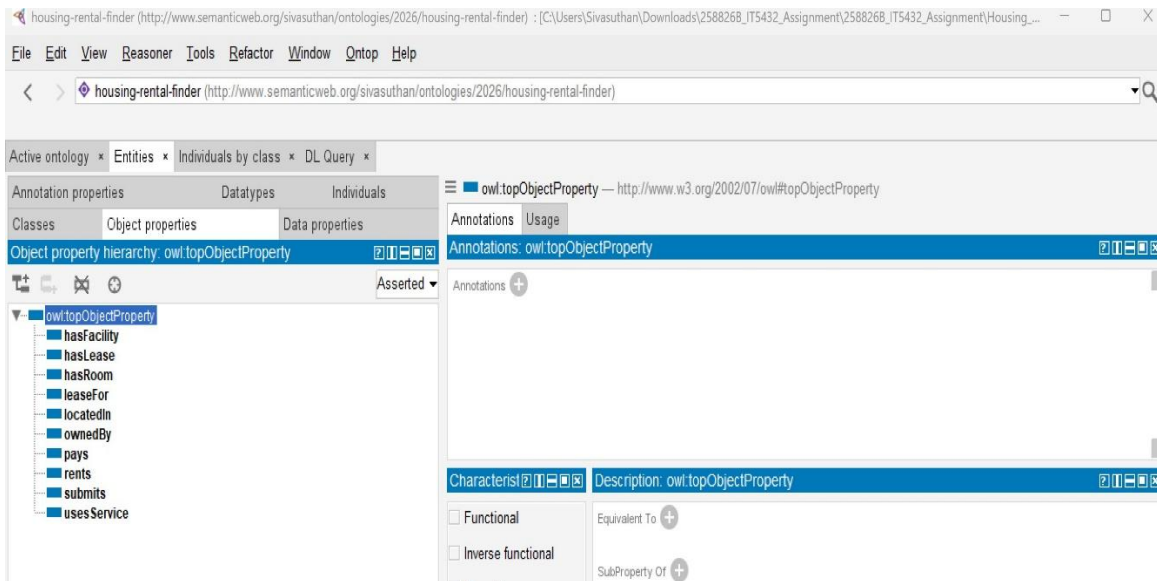
4. Methodology

4.1 Domain Analysis and Ontology Design Main

classes:



Object properties:



4.2 Ontology Population

The ontology was populated with:

- Properties in Colombo, Kandy, and Galle
- Tenant categories
- Lease records
- Landlords and ownership relationships

4.3 Semantic Data Loading

The ontology file is loaded using RDFLib.

4.4 Semantic Query Processing

SPARQL is used to retrieve properties and cities.

4.5 Result Filtering and Ranking

Results are filtered by city and tenant type.

Relevance scores are assigned:

+2 for city match

+2 for tenant match

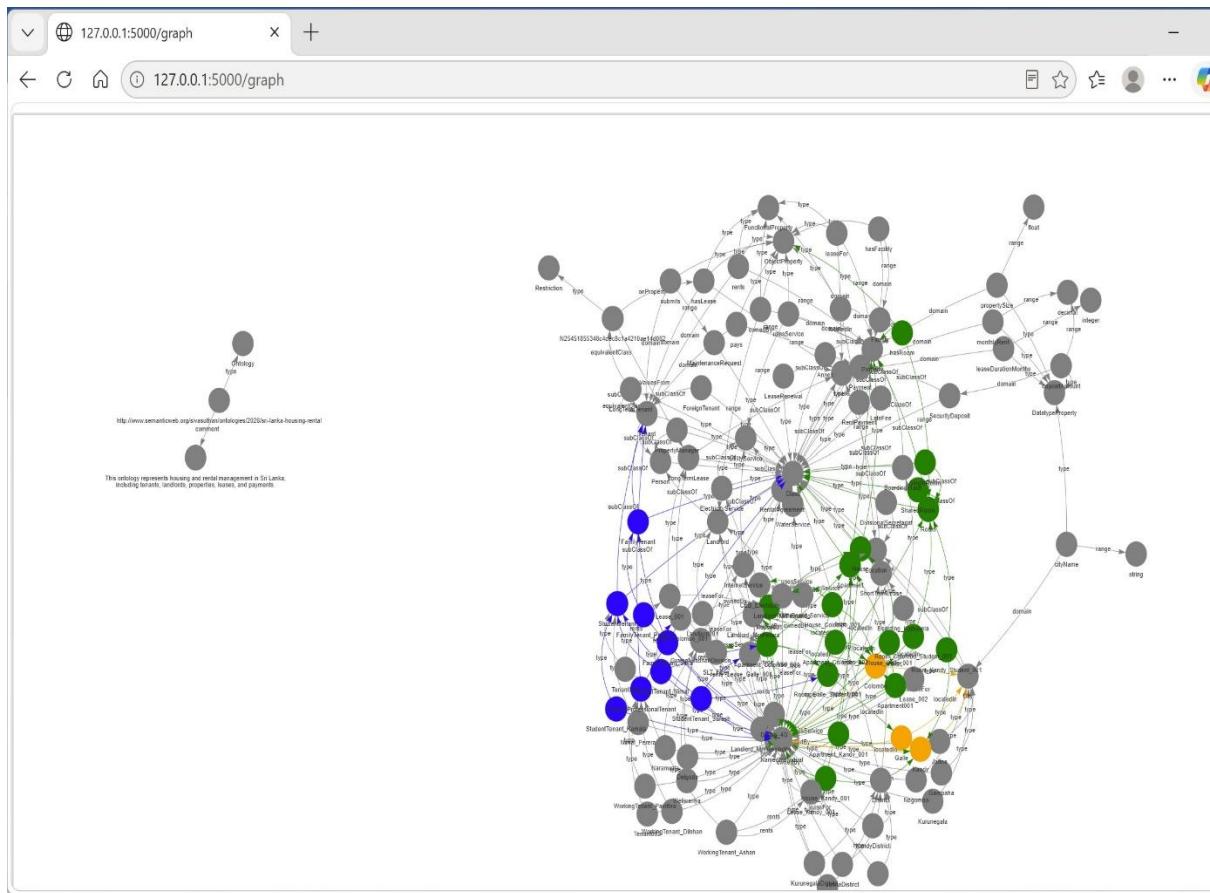
Results are ranked and best choice highlighted.

4.6 Knowledge Graph Generation

RDF triples are converted into a graph using PyVis.

Nodes are color-coded:

- Cities → Orange
- Properties → Green
- Tenants → Blue



4.7 Web Application Integration

Flask integrates all components.

Routes:

- / → Search interface
- /graph → Knowledge graph

GitHub Link:

<https://github.com/Suthan99/Ontology-Based-Smart-Housing-Rental-Finder-for-Sri-Lanka.git>